

Paper I — Low-redshift observational closure of the BPB/MBE branch-projection dictionary

Version v0.4.1: HAL corpus release hotfix

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Abstract

We test the low-redshift observational closure of the BPB/MBE branch-projection dictionary derived in Paper 0. This is Paper I of the BPB/MBE corpus release: Paper 0 defines the dictionary, while Papers I–VII test its descendants across low-redshift cosmology, galaxies, CMB, compact objects and projected Weyl probes. The dictionary fixes the rounded surviving branch anchor

$$s_{\text{surv},0} = 0.800$$

and predicts a negative effective growth drag

$$\beta_{\text{grow}} \simeq -0.2067.$$

In the exact MAP1782 artefact used here, these become

$$s_{\text{surv},0} = 0.7956924397708606, \quad \beta_{\text{grow}} = -0.2106621406010265.$$

We propagate this dictionary into the late-time background, matter-growth and Weyl/lensing sectors, and confront the resulting cosmology with SNe, cosmic chronometers, DESI DR2 BAO, redshift-space growth, a compressed KiDS S_8 constraint and ACT DR6 lensing. A matched Λ CDM comparator is evaluated on the same data stack. The clean BPB/MBE full-joint solution reaches

$$\chi_{\text{BPB}}^2 = 1782.4420773237218,$$

whereas the matched Λ CDM comparator gives

$$\chi_{\Lambda\text{CDM}}^2 = 1798.2712190366963,$$

corresponding to

$$\Delta\chi^2 \equiv \chi_{\text{BPB}}^2 - \chi_{\Lambda\text{CDM}}^2 = -15.829141712974518.$$

The improvement is obtained mainly through the background, growth and KiDS-amplitude blocks; the ACT block is a Weyl-sector stress test, not the block in which the final full-joint BPB/MBE solution beats Λ CDM. The ACT-only diagnostic nevertheless favours a positive Weyl-auto closure and rejects the negative E_G -like closure. The central physical distinction of the paper is therefore

$$\boxed{\mu_m \neq \Sigma},$$

where μ_m controls matter growth and Σ controls Weyl/lensing response. In this v0.4.1 corpus-release version, the v0.3 numerical lock is preserved and connected to the release-level manifest, claims index and reviewer route. AIC and BIC robustness are both reported: AIC remains favourable even with $\Delta k = 2$, and the BIC bracket remains favourable over the relevant compressed ACT-lensing range.

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1 Executive lock

This v0.4.1 corpus-release draft preserves the v0.3 numerical lock and connects the Paper-I low-redshift result to the release-level artefacts. The numerical content is extracted from `cosmology_outputs.zip` and `map1782_early_galaxies.yaml`. The locked numerical anchors are:

Quantity	Value	Status
χ_{BPB}^2	1782.4420773237218	locked
$\chi_{\Lambda\text{CDM}}^2$	1798.2712190366963	locked
$\Delta\chi^2$, BPB – ΛCDM	-15.829141712974518	locked
rounded $s_{\text{surv},0}$	0.800	Paper 0 prose anchor
exact $s_{\text{surv},0}$	0.7956924397708606	MAP1782 diagnostic
rounded β_{grow}	-0.2067	Paper 0 prose anchor
exact β_{grow}	-0.2106621406010265	full-joint MAP

The sign convention is fixed throughout:

$$\Delta\chi^2 = \chi_{\text{BPB}}^2 - \chi_{\Lambda\text{CDM}}^2,$$

so negative values favour BPB/MBE.

2 Corpus placement and release navigation

This manuscript is the low-redshift observational member of the BPB/MBE corpus release. Its role is deliberately narrower than Paper 0 and narrower than the later probe-specific papers. Paper 0 derives the branch-projection dictionary and the shared numerical anchors. Paper I asks whether those anchors survive a fair low-redshift data stack when compared to a matched ΛCDM optimizer. The later papers then test different descendants of the same dictionary.

Release artefact	Role for Paper I
00_README_CORPUS.pdf	Human entry point: explains how Paper I fits between the theoretical dictionary and the other observational descendants.
00_MANIFEST_CORPUS.csv	Machine-readable corpus map: identifies Paper I as the low-redshift joint-likelihood gate.
00_CLAIMS_INDEX.csv	Claim-level source map: links the Paper-I claims to the numerical locks and guardrails used here.
00_REVIEWER_ROUTE.md	Reviewer route: indicates the fastest way to falsify or validate the low-redshift claim.

The release-level reading order is therefore:

$$00_README_CORPUS \rightarrow \text{Paper 0} \rightarrow \text{Paper I} \rightarrow \text{probe-specific descendants}.$$

Paper I can be read independently as a low-redshift model-comparison and diagnostic paper, but its parameters should not be interpreted as free phenomenological late-time parameters detached from Paper 0. They are imported dictionary descendants, with the exact-versus-rounded conventions stated explicitly below.

Paper-I guardrail in the corpus. The allowed Paper-I claim is a MAP-level, source-locked low-redshift closure: the BPB/MBE dictionary improves the matched ΛCDM comparator on the specified joint stack, with the blockwise origin of the gain exposed. The forbidden claim is a final Bayesian evidence result, a full Planck-primary-CMB closure, or a proof that the low-redshift gate alone establishes the entire corpus.

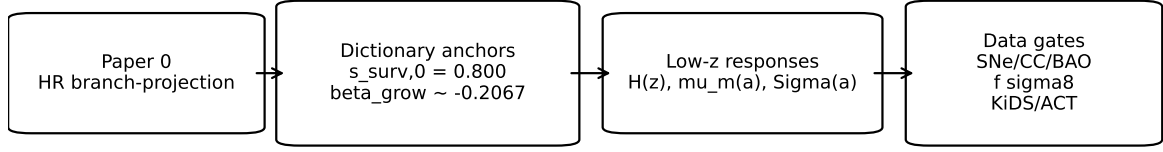


Figure 1: Dictionary flow used in Paper I. Paper 0 supplies the HR branch-projection dictionary and the rounded anchors $s_{\text{surv},0} = 0.800$ and $\beta_{\text{grow}} \simeq -0.2067$. Paper I tests the resulting background, matter-growth and Weyl/lensing responses against low-redshift data.

3 Scientific framing

Paper 0 established the HR branch-projection dictionary. Its purpose was to define the map from the biface branch structure to the effective late-time cosmological sector. The present paper asks the first observational question:

Does the Paper 0 dictionary survive the standard low-redshift cosmological probes when confronted with a fair Λ CDM comparator?

The low-redshift probes do not test a single observable. SNe and BAO constrain distances. Cosmic chronometers constrain $H(z)$. Redshift-space distortions constrain matter growth through $f\sigma_8$. KiDS and ACT constrain a Weyl/lensing response. A successful low-redshift closure must therefore pass three gates:

- (G1) **Background gate:** the branch-projection background must not spoil SNe, CC or BAO.
- (G2) **Growth gate:** the effective growth response must be compatible with $f\sigma_8$.
- (G3) **Weyl/lensing gate:** the lensing response must remain compatible with KiDS-like low-redshift shear information and ACT DR6 CMB lensing.

The key point is that these gates are not identical. In particular, BPB/MBE naturally separates the matter response from the Weyl response. We therefore do not impose the naive identification

$$\mu_m = \Sigma.$$

Instead, μ_m controls the matter-growth sector, while Σ controls the Weyl/lensing sector.

4 Paper 0 dictionary imported into the low-redshift sector

4.1 Branch-projection anchors

The current corpus-release manuscript uses two numerical conventions:

- In prose, the Paper 0 rounded dictionary anchor is

$$s_{\text{surv},0} = 0.800.$$

- In exact diagnostic tables, the MAP1782 YAML value is

$$s_{\text{surv},0} = 0.7956924397708606.$$

This is an exact-versus-rounded distinction, not a contradiction; the same convention is used in the corpus manifest and claims index.

The corresponding exact branch diagnostics from `map1782_early_galaxies.yaml` are:

Diagnostic	Value
$\Omega_{m,\text{eff},0}$	0.3570952870771213
$\Omega_{m,\text{surv},0}$	0.2841380202050705
$\Omega_{m,\text{ind},0}$	0.0729572668720508
$\Omega_{\text{de},\text{eff},0}$	0.6429047129228787
$s_{\text{surv},0}$	0.7956924397708606
u_0	0.22854255
$J_{\text{bf},0}$	0.0

The growth anchor is similarly reported at two levels:

$$\beta_{\text{grow}} \simeq -0.2067$$

in the Paper 0 rounded language, and

$$\beta_{\text{grow}} = -0.2106621406010265$$

in the exact MAP table.

4.2 Background sector

The background sector is evaluated through

$$H(z) = H_{\text{BPB}}(z; \theta_{\text{bg}}),$$

with

$$D_H(z) = \frac{c}{H(z)}, \quad D_M(z) = c \int_0^z \frac{dz'}{H(z')},$$

and

$$D_V(z) = \left[z D_M^2(z) D_H(z) \right]^{1/3}.$$

The BAO observables are ratios relative to the drag scale r_d :

$$D_M/r_d, \quad D_H/r_d, \quad D_V/r_d.$$

4.3 Matter-growth response

The matter-growth sector is represented schematically by

$$\delta_m'' + \mathcal{F}(a)\delta_m' - \frac{3}{2}\Omega_m(a)\mu_m(a)\delta_m = 0,$$

where μ_m is the effective matter response. The observed quantity is

$$f\sigma_8(z) = f(z)\sigma_8(z), \quad f(z) = \frac{d \ln D}{d \ln a}.$$

The negative β_{grow} inherited from the dictionary is not a nuisance amplitude; it is the dictionary-level control of late-time growth response.

4.4 Weyl/lensing response

The full-joint MAP used in this paper fixes a positive Weyl-closure coefficient

$$c_{\Sigma} = 0.71$$

in the ACT-compatible run. The baseline convention for the main Paper I run is the linear Weyl-amplitude response

$$\Sigma(a) = 1 + c_{\Sigma} u(a),$$

where $u(a)$ is the branch-projection memory kernel.

The apparently “quadratic” entry in the ACT diagnostic is not a different baseline function $\Sigma(a)$. It is the expected Weyl auto-power response: a lensing auto-spectrum scales as Σ^2 when the field-amplitude response is Σ . Thus this corpus-release draft freezes the convention:

The baseline closure is linear in $u(a)$ at the Weyl-amplitude level; ACT auto-power diagnostics may use Σ^2 because the observed spectrum is an auto-correlation.

This status is deliberately conservative: c_{Σ} is fixed to 0.71 in the current MAP artefact, but is not yet claimed as a strict Paper 0 prediction. Operationally, $c_{\Sigma} = 0.71$ is selected by the ACT Weyl-closure diagnostic reported in Section 10; it is fixed in the full-joint artefact rather than silently re-profiled inside the main low-redshift comparison. For AIC/BIC accounting, we therefore count it as an additional effective parameter in the conservative scenario. The diagnostic ACT-only scan is reported separately in Section 10; it is not silently substituted for the full-joint baseline.

5 Data stack and fair comparator

The full-joint stack is:

$$\text{SNe} + \text{CC} + \text{DESI DR2 BAO} + f\sigma_8 + \text{KiDS } S_8 + \text{ACT DR6 lensing}.$$

The SNe block is Pantheon+-like in size, with 1701 light curves as in the Pantheon+ release and cosmology analysis [Scolnic et al.(2022), Brout et al.(2022)]. The BAO block follows the DESI DR2 BAO release [Abdul Karim et al.(2025)]. The KiDS compressed amplitude is tied to the KiDS-1000 cosmic-shear analysis [Asgari et al.(2021)]. The ACT block uses the ACT DR6 CMB-lensing likelihood family associated with the DR6 lensing map and power-spectrum analyses [Qu et al.(2024), Madhavacheril et al.(2024)].

The matched Λ CDM comparator satisfies the following protocol:

1. Same data stack.
2. Same covariance and cuts, as encoded in the run artefacts.
3. Same optimisation style at the MAP level.
4. Same blockwise reporting.
5. No comparison against an externally frozen or unoptimised Λ CDM point.

6 MAP parameter tables

6.1 BPB/MBE full-joint MAP

Baseline file: `bpb_joint_lowz_kids_act_map_clean_rerun_c071.json`.

Quantity	Value
H_0	68.75605125946288
ω_m physical	0.16881303263838016
ω_b	0.022361947526776345
a_{bg}	0.40323595509059257
Δ_{bg}	0.0040466091922042036
sr_{max} / split relic	0.10963720865137937
β_{grow}	-0.2106621406010265
$\sigma_{8,0}$	0.803106830968733
z_{on}	2.075180860326095
fixed transition width dz	0.3
r_d [Mpc]	140.85340478649732
S_8 model	0.7573060908558676
$\Sigma(z=0)$	1.0706990140944435
$\Sigma(z=2)$	1.0440273339970776
$\Sigma(z=4)$	1.000000189012452

Sigma bookkeeping. The three $\Sigma(z)$ entries in this table are read directly from the full-joint MAP artefact. They should not be recomputed using the $u(z)$ values reported later in the ACT-only diagnostic table, because those $u(z)$ values come from a separate diagnostic artefact and not from the full-joint MAP table. In v0.3 the rule is therefore strict: whenever $\Sigma = 1 + c_\Sigma u$ is evaluated numerically, Σ , u and c_Σ must be taken from the same artefact.

6.2 Matched Λ CDM full-joint MAP

Baseline file: `lcdm_joint_lowz_kids_act_map_clean_rerun.json`.

Quantity	Value
H_0	67.64647124816678
ω_m	0.14736194442946995
ω_b	0.02208260300511729
$\sigma_{8,0}$	0.8172958083274081
h	0.6764647124816677
Ω_m	0.32202904932155235
Ω_b	0.048256961319390994
r_d [Mpc]	146.02554947207477
S_8 model	0.7209207529497804

7 Blockwise full-joint result

The blockwise comparison is now fully specified.

Block	BPB χ^2	Λ CDM χ^2	$\Delta\chi^2$ BPB – Λ CDM
SNe	1728.5881669819355	1732.8501538150012	-4.2619868330657
CC	13.00117044274172	14.781481598568028	-1.780311155826308
BAO	9.054477646231566	12.228096849629807	-3.173619203398241
$f\sigma_8$	13.148459383821173	21.106078538971023	-7.95761915514985
KiDS S_8	0.004981472549609801	2.5174115554021825	-2.512430082852573
ACT DR6 lensing	18.64482139644229	14.787996679124046	+3.856824717318244
Total	1782.4420773237218	1798.2712190366963	-15.829141712974518

This blockwise table is the core v0.1 correction. It prevents the misleading statement that ACT is the final full-joint winning block. The correct statement is:

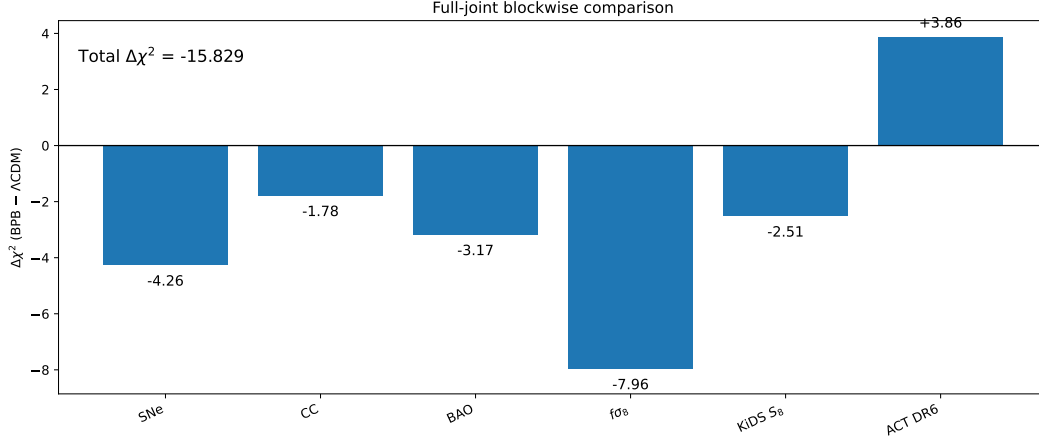


Figure 2: Full-joint blockwise $\Delta\chi^2 = \chi^2_{\text{BPB}} - \chi^2_{\Lambda\text{CDM}}$. BPB/MBE wins the total through the SNe, CC, BAO, $f\sigma_8$ and KiDS-amplitude blocks. The ACT block is positive, meaning that ACT remains a stress gate rather than the block in which the final full-joint BPB/MBE solution beats Λ CDM.

BPB/MBE wins the full joint comparison by the background, growth and KiDS-amplitude blocks; ACT is a Weyl-sector stress test that the model survives with the positive $c_\Sigma = 0.71$ closure.

8 Incremental dataset-stack ladder

The available artefacts provide a dataset-stack ladder. This is not a perfect same-data ablation table, because each row corresponds to a different data stack and a separately optimised MAP. It is nevertheless useful as a transparent build-up of the result.

Stack	Model	Total χ^2	File
low-z only: SNe+CC+BAO+ $f\sigma_8$	BPB	1765.3094254524597	bpb_joint_lowz_map_clean.json
low-z only: SNe+CC+BAO+ $f\sigma_8$	Λ CDM	1772.592893548119	lcdm_joint_lowz_map_clean.json
low-z+KiDS S_8	BPB	1763.7190116796417	bpb_joint_lowz_kids_map_clean.json
low-z+KiDS S_8	Λ CDM	1781.9281083558033	lcdm_joint_lowz_kids_map_clean.json
low-z+KiDS S_8 +ACT, $c_\Sigma = 0.71$	BPB	1782.4420773237218	bpb_joint_lowz_kids_act_map_clean_rerun_c071.json
low-z+KiDS S_8 +ACT	Λ CDM	1798.2712190366963	lcdm_joint_lowz_kids_act_map_clean_rerun.json

The corresponding pairwise improvements are:

Stack	$\Delta\chi^2$ BPB - Λ CDM
low-z only	-7.283468095659368
low-z+KiDS S_8	-18.2090966761616
low-z+KiDS S_8 +ACT	-15.829141712974518

A controlled same-data ablation ladder would still be useful for a later draft, but it is no longer a hidden placeholder. The current paper can honestly publish the available ladder as an incremental stack comparison.

9 Same-data ablation status

This manuscript separates three different objects that must not be conflated:

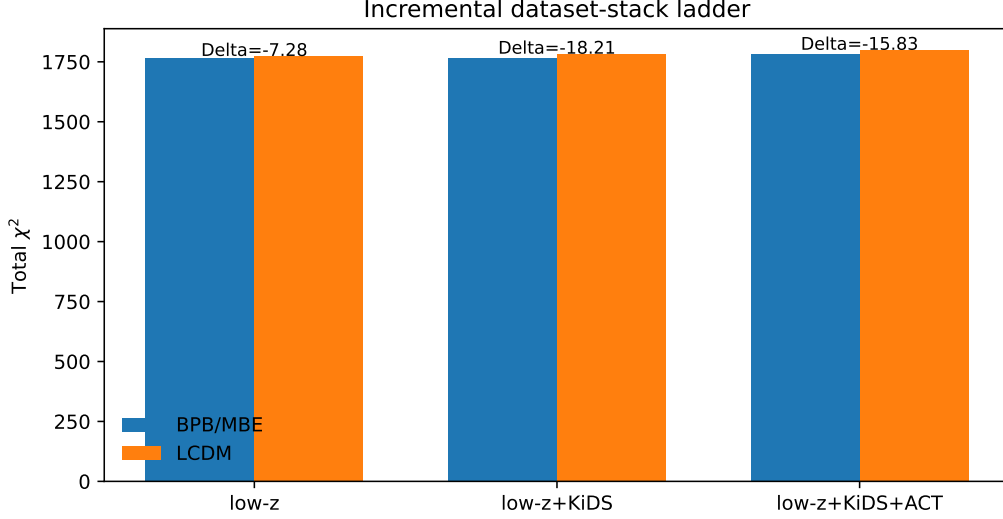


Figure 3: Incremental dataset-stack ladder from the available MAP files. The low- z stack already favours BPB/MBE by $\Delta\chi^2 \simeq -7.28$. Adding compressed KiDS strengthens the preference. Adding ACT reduces the net gain relative to low- z +KiDS but leaves the full-joint result at $\Delta\chi^2 \simeq -15.83$.

1. The **full-joint blockwise table** is a same-data comparison between BPB and Λ CDM on the final stack.
2. The **incremental stack ladder** is not a same-data ablation; it changes the data stack and is labelled as such.
3. The **ACT Weyl-closure diagnostic** is a same-ACT-likelihood ablation of the Weyl response at the frozen point.

The uploaded artefacts do not contain a global controlled same-data ablation in which the full final data vector is held fixed while BPB components are switched on and off one by one. This corpus-release version therefore does not claim such a result. Instead, it makes the limitation explicit and uses only artefact-backed comparisons.

10 ACT/Weyl diagnostic

The ACT-only diagnostic is reported separately from the full-joint comparison. It addresses the question:

What Weyl closure does ACT prefer at the MAP1782 seed point?

The source files are:

```
act_sigma_power_diagnostic_seed01.json
scan_act_bpb_csigma_refined_on_frozen_round1_act_baseline.json
```

ACT diagnostic case	c_Σ	power	ACT χ^2
no Σ response	0.0	0	27.654787887022678
linear ACT closure	0.71	1	21.365956457070062
quadratic ACT closure	0.71	2	19.122586898896895
E_G -like linear closure	-3.3	1	110.11470093349499
E_G -like quadratic closure	-3.3	2	219.10242541490783

The refined scalar scan around the positive ACT optimum gives:

1. Full-joint blockwise	same data stack	resolved
2. Incremental stack ladder	different data stacks	resolved
3. ACT Weyl closure	same ACT likelihood	resolved
4. Global same-data ablation	requires new runs	not claimed

v0.2 rule: publish only resolved artefact-backed comparisons; define but do not claim missing global ablation.

Figure 4: Ablation status in v0.2. The paper keeps resolved, artefact-backed comparisons separate from the still-missing global same-data ablation. This prevents the stack ladder from being overinterpreted as a component-only ablation.

Quantity	Value
best c_Σ	0.7100000000000001
BPB ACT χ^2 , bespoke closure	19.122586898896895
BPB ACT χ^2 , no/minimal reference	27.128611029395095
Λ CDM native ACT χ^2 in this diagnostic	22.9270917814448
$\Delta\chi^2$, bespoke – effective reference	-8.0060241304982
$\Delta\chi^2$, bespoke – Λ CDM native	-3.8045048825479064
$u(z=0)$	0.09722304639511888
$u(z=2)$	0.06813888382070675
$u(z=4)$	0.0000003689030393947221
$\Sigma(z=0)$	1.0690283629405344
$\Sigma(z=2)$	1.0483786075127017
$\Sigma(z=4)$	1.0000002619211579

Diagnostic-only status of these Σ values. The $u(z)$ and $\Sigma(z)$ values in this table belong to the ACT-only closure diagnostic. They are internally consistent with the diagnostic scan, but they are not the same entries as the full-joint MAP $\Sigma(z)$ values quoted in the main MAP table. The small numerical offsets between the two sets of $\Sigma(z)$ values therefore trace artefact-level bookkeeping, not a change in the scientific claim. The publication claim remains the same: the final full-joint run uses a positive Weyl closure, and the ACT diagnostic shows that positive Weyl-auto response is favoured over either no closure or the negative E_G -like closure.

The diagnostic result supports the Paper I statement that matter growth and Weyl/lensing are separate responses. It does not license the stronger claim that the ACT block alone is the winning full-joint block.

11 AIC and parameter-count lock

For AIC,

$$\text{AIC} = \chi^2 + 2k,$$

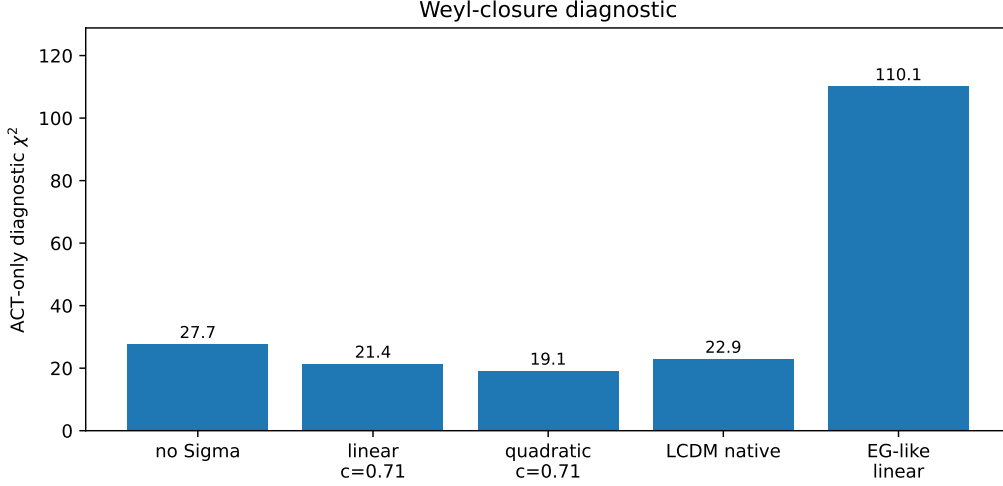


Figure 5: ACT-only Weyl-closure diagnostic. ACT strongly rejects the negative E_G -like closure and prefers a positive Weyl-auto response. This supports the structural claim $\mu_m \neq \Sigma$, but it is not the same statement as the final full-joint blockwise comparison.

so

$$\Delta\text{AIC} = \Delta\chi^2 + 2\Delta k.$$

Using

$$\Delta\chi^2 = -15.829141712974518,$$

we obtain:

Scenario	Δk	ΔAIC
BPB has one additional effective parameter	1	-13.829141712974518
BPB has two additional effective parameters, counting c_Σ	2	-11.829141712974518

Thus, even under conservative counting in which c_Σ is treated as an additional effective Weyl-closure parameter, the AIC comparison remains favourable to BPB/MBE.

For BIC,

$$\text{BIC} = \chi^2 + k \ln N,$$

so

$$\Delta\text{BIC} = \Delta\chi^2 + \Delta k \ln N.$$

The artefact `canonical_dataset_summary.json` fixes the non-ACT data-vector size:

$$N_{\text{SNe}} = 1701, \quad N_{\text{CC}} = 32, \quad N_{\text{BAO}} = 9, \quad N_{f\sigma_8} = 22.$$

The KiDS block is the compressed S_8 gate, counted as one point in this Paper I likelihood. Thus

$$N_{\text{nonACT}} = 1701 + 32 + 9 + 22 + 1 = 1765.$$

The uploaded artefacts do not expose the internal ACT bandpower count as a standalone integer. Therefore v0.3 reports the BIC as a robustness bracket with

$$N = 1765 + N_{\text{ACT,eff}}.$$

For any reasonable ACT compressed lensing-vector size, the conclusion is unchanged. At $N_{\text{ACT,eff}} = 0$, one obtains

$$\Delta\text{BIC}(\Delta k = 1) = -8.353,$$

$$\Delta\text{BIC}(\Delta k = 2) = -0.877.$$

Even for $N_{\text{ACT,eff}} = 100$, the values remain approximately -8.30 and -0.77 . The $\Delta k = 2$ BIC case would flip sign only if $N \gtrsim 2737$, which would require about 972 additional effective ACT points beyond the non-ACT vector. This is not the regime represented by the ACT lensing block used here.

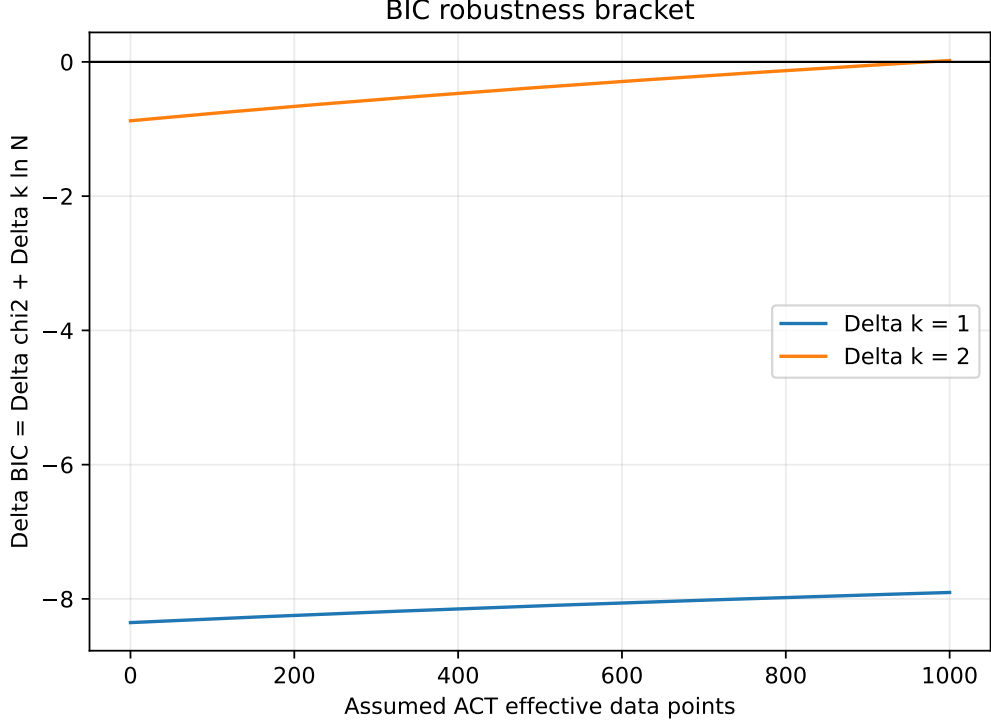


Figure 6: BIC robustness bracket as a function of the assumed ACT effective data-vector size. For both accepted parameter-count scenarios, $\Delta k = 1$ and $\Delta k = 2$, the BIC remains favourable over the relevant compressed-likelihood range.

12 Main claims

Claim 1: dictionary transport

The HR branch-projection dictionary can be transported into the low-redshift cosmological sector using the Paper 0 rounded anchors

$$s_{\text{surv},0} = 0.800, \quad \beta_{\text{grow}} \simeq -0.2067,$$

with exact MAP1782 values reported in the tables.

Claim 2: background survival

The background gate is passed. The full-joint blockwise improvement in SNe, CC and BAO is

$$(-4.2619868330657) + (-1.780311155826308) + (-3.173619203398241) = -9.215917192290249.$$

Claim 3: growth-sector improvement

The growth gate is passed. The $f\sigma_8$ block gives

$$\Delta\chi^2_{f\sigma_8} = -7.95761915514985.$$

Claim 4: KiDS compressed amplitude support

The compressed KiDS block gives

$$\Delta\chi^2_{\text{KiDS}} = -2.512430082852573.$$

This is an amplitude-level gate, not yet the full KiDS tomographic likelihood.

Claim 5: Weyl response differs from matter response

The ACT diagnostic requires a positive Weyl-auto closure and rejects the negative E_G -like closure. The final full-joint ACT block is not the main source of the total win, but the ACT closure supports the structural distinction

$$\mu_m \neq \Sigma.$$

Claim 6: full-joint MAP improvement

The full-joint improvement is

$$\Delta\chi^2 = -15.829141712974518.$$

Even with a conservative AIC penalty that counts c_Σ as an extra effective parameter, the AIC remains favourable.

Claim 7: corpus-level role

Paper I is the first observational descendant of the Paper 0 dictionary. It validates the low-redshift background-growth-Weyl transport at MAP level, while deliberately leaving the other corpus claims to their own probe-specific papers and source locks.

13 Discussion

13.1 What this paper establishes

This paper establishes that the Paper 0 branch-projection dictionary passes a first low-redshift observational closure test. It does not merely fit one probe: it survives a joint background-growth-lensing stack and improves the matched Λ CDM comparator at the MAP level.

13.2 What this paper does not establish

This paper does not claim:

1. a full Planck primary-CMB likelihood closure;
2. a full KiDS tomographic shear likelihood closure;
3. a SPARC or BTFR galaxy-scale proof;
4. a neutron-star-sector proof;
5. a final Bayesian evidence comparison;
6. a proof that no other modified-gravity model can mimic the same low-redshift response.

13.3 Correct ACT interpretation

The corrected ACT narrative is essential. The final full-joint ACT block is

$$\Delta\chi_{\text{ACT}}^2 = +3.856824717318244,$$

so ACT alone favours the matched Λ CDM block in the final full-joint decomposition. However, the ACT-only Weyl diagnostic shows that BPB/MBE improves substantially when a positive Weyl-auto closure is used, and that the E_G -like negative closure is catastrophic for ACT. The defensible statement is:

ACT is the Weyl-sector stress gate. It supports a positive Weyl-auto closure and enforces $\mu_m \neq \Sigma$, but the final full-joint BPB/MBE improvement is carried by background, growth and KiDS-amplitude blocks.

14 v0.4.1 corpus-release hotfix summary

The v0.4.1 hotfix is a corpus-release patch on top of v0.4. It does not change the numerical result locked in v0.3. It adds the release-navigation layer, connects Paper I to `00_README_CORPUS.pdf`, `00_MANIFEST_CORPUS.csv`, `00_CLAIMS_INDEX.csv` and `00_REVIEWER_ROUTE.md`, and states explicitly that Paper I is the low-redshift observational gate of the BPB/MBE corpus. The v0.3 audit fixes are retained: the BIC sign-flip threshold for the conservative $\Delta k = 2$ case remains $N_{\text{crit}} \simeq 2737$, and the full-joint MAP $\Sigma(z)$ values remain separated from the ACT-only diagnostic $u(z)$ and $\Sigma(z)$ values. No locked value of $\Delta\chi^2$, AIC/BIC, the blockwise comparison, or the core claim $\mu_m \neq \Sigma$ is changed. The v0.4.1 hotfix also clarifies in Section 4.4 that $c_\Sigma = 0.71$ is selected by the ACT Weyl-closure diagnostic and regenerates Figure 2 with the explicit label $\Delta\chi^2$.

15 Conclusions

1. We tested the low-redshift observational closure of the BPB/MBE branch-projection dictionary derived in Paper 0.
2. The rounded Paper 0 anchors are $s_{\text{surv},0} = 0.800$ and $\beta_{\text{grow}} \simeq -0.2067$.
3. The exact MAP1782 values used in the numerical tables are $s_{\text{surv},0} = 0.7956924397708606$ and $\beta_{\text{grow}} = -0.2106621406010265$.
4. The clean full-joint BPB/MBE run gives $\chi_{\text{BPB}}^2 = 1782.4420773237218$.
5. The matched ΛCDM comparator gives $\chi_{\Lambda\text{CDM}}^2 = 1798.2712190366963$.
6. The resulting improvement is $\Delta\chi^2 = -15.829141712974518$.
7. The improvement is driven by SNe, CC, BAO, $f\sigma_8$ and compressed KiDS S_8 .
8. The final full-joint ACT block alone is not a BPB win: $\Delta\chi_{\text{ACT}}^2 = +3.856824717318244$.
9. ACT nevertheless supplies a crucial Weyl-sector diagnostic: it prefers positive Weyl-auto closure and rejects the negative E_G -like closure.
10. The central physical conclusion is $\mu_m \neq \Sigma$.
11. Even counting c_Σ as an extra effective parameter, the AIC remains favourable: $\Delta\text{AIC} = -11.829141712974518$.
12. v0.4.1 keeps the v0.3 BIC robustness bracket and the $\Sigma(a) = 1 + c_\Sigma u(a)$ amplitude convention, while adding the corpus-release navigation and claim/source-lock guardrails. The remaining non-numerical open items are a future global controlled same-data ablation run and production-grade reproduction scripts for the final figure set.

A Corpus-release source pointers

The Paper-I source package includes a `release_index/` directory containing the stage-1 corpus index files used during this release preparation. These files are not inputs to the low-redshift likelihood calculation; they are navigation and provenance artefacts.

File	Function
<code>release_index/00_MANIFEST_CORPUS.csv</code>	one-row-per-paper corpus manifest
<code>release_index/00_CLAIMS_INDEX.csv</code>	claim/source-lock index parsed from the available outputs
<code>release_index/00_RELEASE_NOTES.md</code>	release-level caveats and packaging notes
<code>release_index/00_REVIEWER_ROUTE.md</code>	fast referee route through the corpus

B Artefact inventory

Key	File and role
BPB full joint	<code>bpb_joint_lowz_kids_act_map_clean_rerun_c071.json</code> . Full-joint BPB/MBE MAP with $c_\Sigma = 0.71$.
LCDM full joint	<code>lcdm_joint_lowz_kids_act_map_clean_rerun.json</code> . Matched full-joint Λ CDM MAP.
BPB low-z	<code>bpb_joint_lowz_map_clean.json</code> . Low-z-only BPB/MBE MAP.
LCDM low-z	<code>lcdm_joint_lowz_map_clean.json</code> . Low-z-only Λ CDM MAP.
BPB low-z+KiDS	<code>bpb_joint_lowz_kids_map_clean.json</code> . Low-z+KiDS BPB/MBE MAP.
LCDM low-z+KiDS	<code>lcdm_joint_lowz_kids_map_clean.json</code> . Low-z+KiDS Λ CDM MAP.
ACT power diag	<code>act_sigma_power_diagnostic_seed01.json</code> . ACT-only closure diagnostic.
ACT csigma scan	<code>scan_act_bpb_csigma_refined_on_frozen_round1_act_baseline.json</code> . Refined ACT scan around $c_\Sigma = 0.71$.
MAP1782 YAML	<code>map1782_early_galaxies.yaml</code> . MAP1782 branch diagnostics for exact $s_{\text{surv},0}$.

C Resolved and open items

Resolved by v0.2:

1. BPB MAP parameter table.
2. Λ CDM MAP parameter table.
3. Full-joint blockwise Λ CDM comparison.
4. Main $\Delta\chi^2$ sign and value.
5. AIC under $\Delta k = 1$ and $\Delta k = 2$.
6. BIC robustness bracket using $N_{\text{nonACT}} = 1765$ and $N = 1765 + N_{\text{ACT,eff}}$.
7. Exact-versus-rounded convention for $s_{\text{surv},0}$ and β_{grow} .
8. ACT diagnostic wording.
9. Baseline Weyl closure convention: $\Sigma(a) = 1 + c_\Sigma u(a)$ at the amplitude level, with Σ^2 in auto-power diagnostics.
10. Bibliography entries for the main external data sets.

Remaining open after v0.2:

1. A future global controlled same-data ablation run, if desired for the referee-proof version.
2. Production-grade reproduction scripts for the final figure set.
3. Final journal-style bibliography formatting.
4. Exact ACT internal bandpower count, if one wants a single-number BIC rather than the robust bracket reported here.

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